

Name:	Mark Ryan T. Pasculado	Date:	August 11, 2026
Course and Section:	BIO 300- B	Instructor:	Mrs. Lilibeth A. Bucol

Genome Exploration II

Basic Genome Structure and Sequence Exploration Using Galaxy

Continuation Assignment • usegalaxy.org • Individual Work

Starting material	The same genome assembly FASTA file used in the previous assignment
Difficulty	Basic / introductory
Main goal	Describe genome structure using simple statistics, filtering, and a small sequence-mining exercise
Expected products	Galaxy History, result table, screenshots, short interpretation, and GitHub documentation

Learning Objectives

- Summarize basic structural properties of a genome assembly.
- Explain N50, L50, genome size, sequence count, and GC content in simple biological terms.
- Inspect how assembly sequences are distributed by length.
- Apply a simple length filter and evaluate how filtering changes an assembly.
- Perform a small open reading frame (ORF) exploration and distinguish an ORF from a confirmed gene.
- Document a reproducible Galaxy analysis in GitHub.

Part 1 — Reconfirm the Genome Identity

Species name	<i>Mus musculus</i>
NCBI Assembly accession	GCF_000001635.27
Assembly level (if known)	Chromosome
FASTA filename in Galaxy	Mus_musculus_genomic.fna
Genome source/database	NCBI RefSeq
Approximate file size	2.8GB

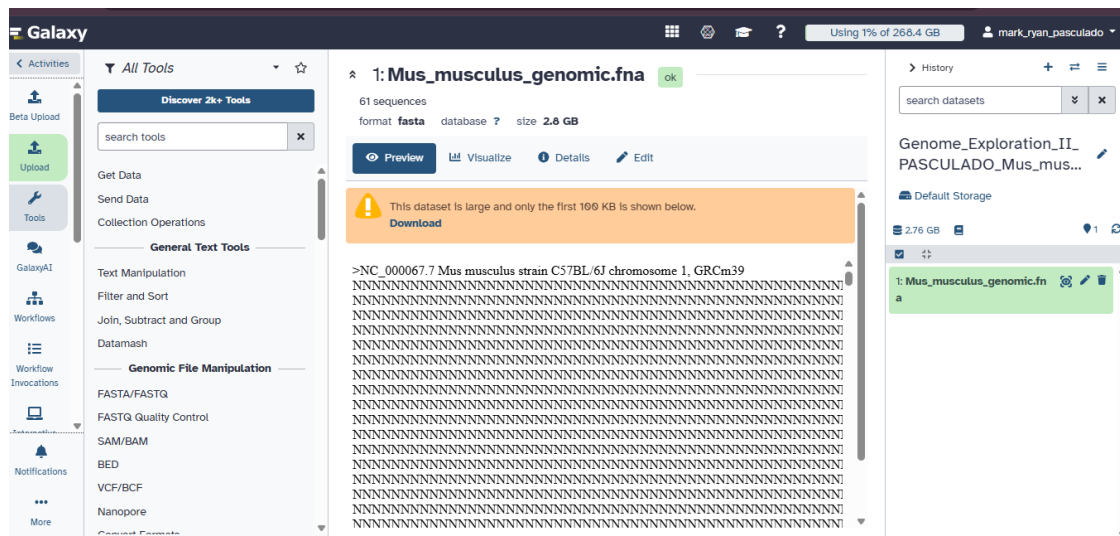


Figure 1. Galaxy History setup and FASTA dataset preview for the house mouse (*Mus musculus*) genome assembly

Part 2 — Generate Basic Assembly Statistics

The *Mus musculus* assembly is a chromosome-level genome assembly with a total scaffold length of 2,728,222,451 bp, distributed across 61 scaffolds. The assembly has a scaffold N50 of 130,530,862 bp and an L50 of 9, meaning that only nine of the largest scaffolds are needed to account for at least half of the total assembly length. The GC content is 41.67%, which represents the proportion of guanine and cytosine bases in the assembly.

Assembly Statistics	Values
Total assembly length (bp or Mb)	2,728,222,451 bp
Number of sequences/contigs/scaffolds	61
Minimum sequence length	1,976 bp
Maximum sequence length	195,154,279 bp
Mean or average sequence length	44,724,958.21 bp
N50	130,530,862 bp
L50, if provided	9
GC content (%)	41.67%

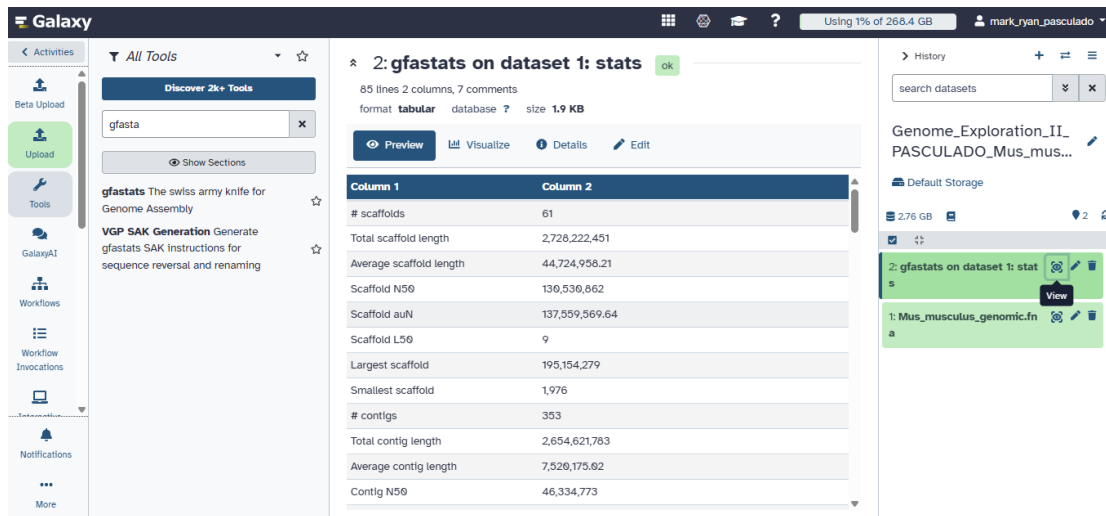


Figure 2. Galaxy summary statistics output for the *Mus musculus* assembly

Part 3 — Examine Sequence-Length Structure

The assembly is highly continuous and is composed of several very large scaffolds representing chromosome-scale sequences. The largest scaffold is 195,154,279 bp, while the smallest scaffold is only 1,976 bp. The large difference between these values shows that the scaffolds are not equally sized and that most of the assembly length is concentrated in the larger chromosome-scale scaffolds.

Rank	Sequence ID	Length (bp)
1	NC_000067.7	195,154,279
2	NC_000068.8	181,755,017
3	NC_000086.8	169,476,592
4	NC_000069.7	159,745,316
5	NC_000070.7	156,860,686

The plot shows that the largest scaffolds have substantially greater sequence lengths, with the first few scaffolds reaching approximately 195 Mb, while the scaffold lengths progressively decrease across the remaining sequences. This distribution indicates that the *Mus musculus* assembly is dominated by large chromosome-scale scaffolds, with only a small number of much shorter sequences contributing to the overall genome size.

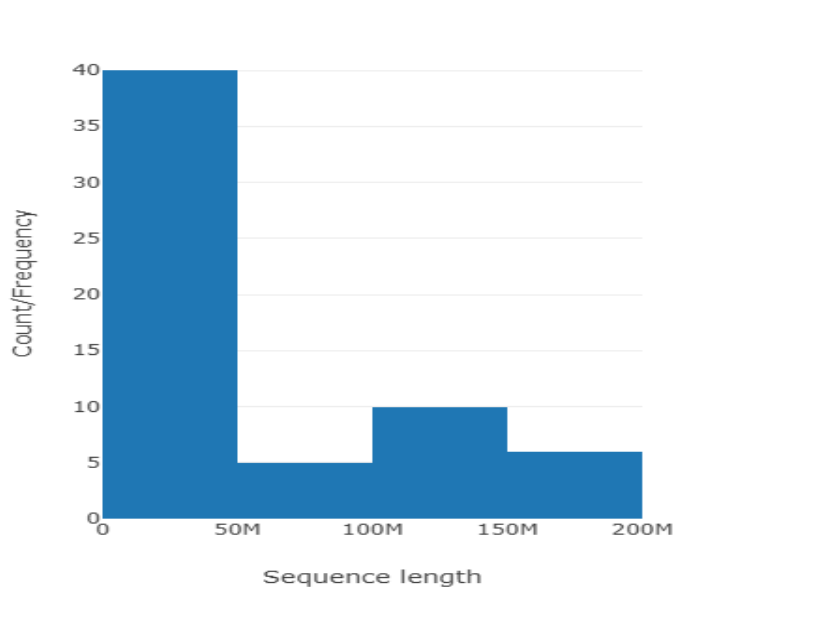


Figure 3. Sequence length distribution across assembled scaffolds in the *Felis catus* genome

Part 4 — Perform a Simple Length-Filtering Experiment

A ≥ 10 kb length filter was applied to the assembly. After filtering, the number of scaffolds decreased from 61 to 59, meaning that two sequences shorter than 10 kb were removed. The total assembly length decreased only slightly, from 2,728,222,451 bp to 2,728,212,063 bp. The N50 remained unchanged at 130,530,862 bp, indicating that the removed short scaffolds had little effect on the overall structure of the assembly.

Metric	Original genome	After ≥ 10 kb filter
Total length	2,728,222,451 bp	2,728,212,063 bp
Number of sequences	61	59
Maximum length	195,154,279 bp	195,154,279 bp
N50	130,530,862 bp	130,530,862 bp
GC content	41.67%	41.67%

How many sequences were removed by the 10 kb filter?

Two sequences were removed, decreasing the number of scaffolds from 61 to 59.

Did the total assembly length decrease greatly or only slightly?

The total assembly length decreased only slightly, by 10,388 bp (about 0.01 Mb). This represents a very small fraction of the approximately 2.73 Gb assembly.

Did N50 change? Why do you think it changed or stayed similar?

The N50 did not change and remained 130,530,862 bp. This is because the removed sequences were very short and did not contribute significantly to the overall assembly length or the calculation of N50.

What does this tell you about the contribution of short sequences to your assembly?

Part 5 — Small ORF Exploration

Metric	Result
Number of ORFs returned	103
Length of the longest ORF	1,494 bp

Answers:

- **How many ORFs were found in the selected sequence/region?**

The analysis identified 103 ORFs in the selected sequence/region. These represent potential coding regions identified based on the sequence characteristics used by the ORF-finding tool.

- **What was the longest ORF?**

The longest ORF was 1,494 bp long. Its length indicates substantial potential for protein-coding capacity, although length alone does not confirm that it is a functional gene.

- **Why should you NOT report every ORF as a real gene?**

An ORF only indicates a sequence with the potential to encode a protein and does not automatically represent a functional gene. Additional evidence, such as sequence similarity, conserved domains, gene expression, evolutionary conservation, or experimental validation, is needed to confirm its biological function.

Part 6 — Short Biological Interpretation

The *Mus musculus* genome assembly is highly continuous, with an approximately 2.73 Gb scaffold length, 130.5 Mb scaffold N50, and L50 of 9, indicating that much of the genome is represented by large scaffolds. Its 41.67% GC content provides a basic measure of nucleotide composition.

Filtering out scaffolds shorter than 10 kb reduced the assembly by only 10,388 bp while leaving the N50 unchanged, showing that these short scaffolds had minimal influence on overall assembly continuity. This suggests that the assembly is largely composed of long, well-assembled genomic sequences.

The ORF analysis identified 103 possible ORFs, with the longest measuring 1,494 bp. These should be considered candidate coding regions rather than confirmed genes, since ORF detection alone does not demonstrate gene expression or biological function.